

PHARMACY PRACTICE [BP703T] ASSIGNMENT-1

PART-A

Ques 1 → Define Hospital Pharmacy.

Ans → Hospital Pharmacy is a specialized department within a hospital under the direction of a professionally qualified Pharmacist, which comprises the art, practice and profession of choosing, preparing, storing, compounding and dispensing medicines and medical devices, advising healthcare professionals and patients.

Ques 2 → Differentiate between inpatient and outpatient pharmacy.

Ans → Inpatient : Serves patients admitted to hospital wards, using unit-dose or floor stock systems Coordinated with nurses and Physicians.

Out-Patient : Serves ambulatory (walking) Patient who visit the hospital and return home, dispensing direct Prescription with patient Counselling.

Ques 3 → What is Medication history Interview?

Ans → A Structured interview Conducted by a Pharmacist to gather a complete record of a Patient's current prescription, OTC (over the Counter), and herbal medication use to prevent reconciliation errors.

Ques 4 → Define Adverse Drug Reaction (ADR).

Ans → Any noxious unintended, and undesired response to a drug that occurs at doses normally used in human for prophylaxis diagnosis, or therapy.

Ques 5 → Define Drug Interaction with one Suitable example.

Ans → A Quantifiable alteration in response of one when administered with another drug, Food, alcohol, Smoking or diagnostic tests.

PART-B

Ques 1 → Explain the organization and functions of a hospital pharmacy with a neat organization chart.

Ans → Hospital Pharmacy operates under a structured hierarchy designed to ensure administrative.

c. Toxic effect.

d. Secondary Pharmacological effect.

ii. Unpredictable : Reactions that are unrelated to the drug's primary pharmacology.

eg → Anaphylactic Shock by penicillin allergy

a. Idiosyncrasy.

b. Allergic reaction.

c. Photosensitivity.

d. Intolerance.

Rawlin's Thompson Classification →

Type →

A. Augmented

Extension of Pharmacological action.

eg - hypoglycemia by insulin.

hemorrhage by warfarin.

B. Bizarre

Immune-mediated hypersensitivity reaction.

eg - Anaphylaxis to penicillin allergy.

C. Chronic

Associated with long-term exposure / therapy and Cumulative dose.

eg - Analgesic nephropathy.

iii. General : Patients come for general checkup and medicines.

Ques 3 → Explain the Causes, classification, reporting and prevention of ADR.

Ans → Patient Related factors : Age, Pediatric, renal, hepatic, genetic and pregnancy.

→ Drug-related Factors : narrow therapeutic index, PolyPharmacy, high dosage, Intrinsic toxicity.

→ Systemic / Environmental Factors : medication error, miscommunication during handoff, use of shelf-medication in minor ailments.

→ Availability of drugs without prescription.

* Classification →

i. Predictable : reactions that are an exaggeration of the drug's known primary Pharmacological action occurring at normal therapeutic doses.

eg → Hypoglycemia by insulin.

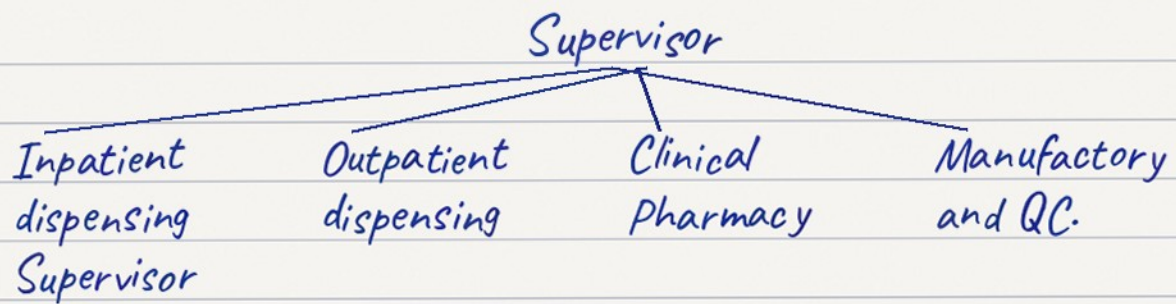
a. Excessive Pharmacological effect

b. Side effect.

- ii. Complete floor Stock : drugs are stored at the nursing station and administered by nurses according to order of physician.
adv → ready availability of drug.
→ Reduce number of Pharmacy personnel
- iii. Combination : Maintains limited inventory of emergency and routine drugs directly on hospital ward while requiring Individual Pharmacist-reviewed Prescription.
Adv. → Provide immediate access.
→ reduce administrative burden.
- iv. Unit Dose Dispensing Method : UD medications are those which are ordered or dispensed In single dose, ready-to-administered packages, delivered in 24 hour carts Containing individual Patient bins.
adv. → better health Services.
→ Reduce medication errors.

OUTPATIENTS.

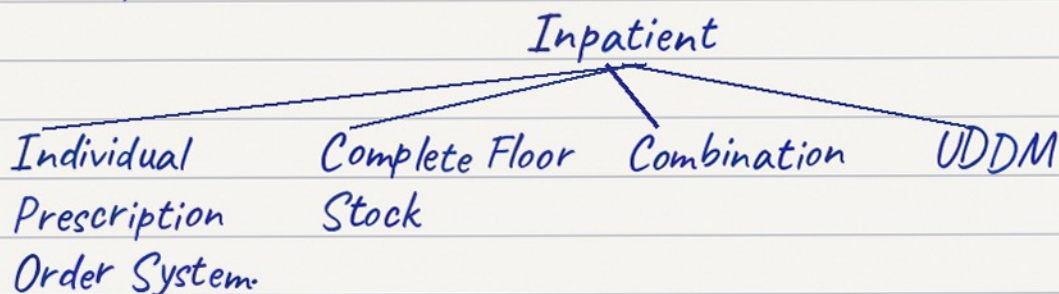
- i. Emergency : 24 hours services are given who requires immediate care.
- ii. Referred : for Patients referred to the hospital for a specific purpose due to lack of facilities.



Functions →

- i. Provide specifications for the purchase of drugs, Chemicals, biologicals etc.
- ii. Proper storing of drug.
- iii. Manufacturing and distribution of medicaments.
- iv. Filling a labelling of drug Containers.
- v. Management of stores.

Ques 2 → Discuss the various drug distribution System Practiced in hospital.



- i. **Individual Prescription :** Prescription Copy is Sent to the pharmacist, dispensing a Supply Specific to that patient.
 adv. → directly reviewed by Pharmacist
 → errors Could be avoided.

- efficiency, accountability and standards.

- i. Pharmacy management director identifies relative Standard organization design molded to match with unique requirements of the pharmacy depending on the number of personnel and Scope of Services oversees overall policies, budgeting and Compliance.
- ii. Chief Pharmacist → manages staff, formulates policies Coordina with other department and P and T Committee.
- iii. Staff Pharmacists → Manage Inpatient and Outpatient dispensing, assist with inventory, Stock maintenance and logistics, Clinical Services.
- iv. Support Staff → Pharmacy technicians.
Storekeepers.



D. Delayed.
Occur Some time after Cessation of treatment
eg - Teratogenicity from thalidomide.

E. End of Use.
Occurs after stoping of drug.
eg - Withdrawl Syndrome.
- rebound hypertension from clonidine. Withdrawl.

F. Failure of therapy
Unexpected failure of drug to produce expected
therapeutic response.
eg - Contraceptive Failure.

Reporting →
Minimal Standard information to be Provided.

- i. Patient information.
- ii. Description of adverse reaction including laboratory results.
- iii. Information related to Suspected drug.
- iv. Reporter information.

Process →
Reporter Should send precise and accurate monitoring in
Standardized format with Sealed envelope and send
directly to pharmacovigilance centre →

Zonal drug information centres.

*

Prevention :-

i.

Take medication history to avoid known allergies and interaction.

ii.

Education patients regarding early warning Signs of toxicity.

iii.

appropriate monitoring required.

iv.

adopting right techniques, doses, route and frequency.

PART-C

Ques 1 → 65 year old male.

Condition → Diabetes and hypertension, knee pain (recent)

Medication → metformin.

glimepiride.

Amlodipine

Aspirin

Diclofenac.

ADR → gastric irritation and hypoglycemia

a. Identify possible medication-related problems

Ans →

a). gastric irritation → due to NSAID's Diclofenac and aspirin prescribed which inhibits protective gastric prostaglandins.

→

hypoglycemia → Due to Sulfonylurea and biguanide Category anti-diabetic drugs glimepiride and metformin leading to low blood glucose.

b. Mention Potential drug Interaction.

Ans → Aspirin and diclofenac increase risk of GIT bleeding, ulceration and renal toxicity.

→ Diclofenac Cause sodium and water retention antagonizing blood - pressure lowering efficacy of antihypertensive drug Amlodipine.

→ Metformin and glimepiride Cause additive effect in hypoglycemic reaction.

c. Suggest Pharmacist Interventions.

Ans → i. Inform Physician about the ADRs.

ii. reviewing the need for diclofenac particularly with diclofenac.

iii. Blood glucose monitoring.

d. Recommend Patient Counselling.

Ans → educate Patients on early Sign of hypoglycemia and gastric Irritation.

→ Advise the patient to avoid Self-medication

→ Take medicines exactly as prescribed.

e. Explain how these adverse events Can be prevented.

Ans → Conducting thorough medication history to flag Contraindications.

→ Eliminating unnecessary Concurrent use of polypharmacy.

→ Regular routine Clinical monitoring.